Research Proposal: Calibrated Selective Automation for Code-Switched Bangla E-Commerce Messaging on a Zero Budget

Author name to be filled
Order Khata project; prepared with an AI coding harness (disclosure to follow venue policy)

17 October 2026 — draft 1

Abstract

Small online shops in Bangladesh receive customer messages in Bangla, romanised Bangla ("Banglish") and English, and must decide which messages a system may answer alone and which require a person. Order Khata is a 48-hour build of such a triage system under a strict zero-cost constraint: one free container, CPU only, no paid API. This proposal argues that the build contains the skeleton of a publishable applied study, but not yet the study itself. The material supports three candidate contributions: (i) a full selective-automation analysis of a shipped policy — 53.6% coverage at 94.2% auto-slice accuracy at $\tau = 0.85$, with a measured structural ceiling of 56.2%; (ii) an empirical result on bounded LLM deferral — when the local model and the LLM disagree at confidence ≥ 0.90, the local model is right 96% of the time against the LLM's 23%, which justifies a tie-breaker-only-when-uncertain design; and (iii) a cost–quality comparison of a 12 ms linear model, a 322M-parameter checkpoint, and an LLM API as first-class systems. The proposal lists the six experiments, the statistics, and the human verification that separate the current artifact from a Q1 submission, with an eight-week plan and a venue ladder.

1 Context and motivation

Order Khata reads customer messages, classifies them into six classes (order, price question, delivery problem, complaint, scam, other), and routes each message to one of three lanes: auto-handled, human review, or scam. The routing decision is a five-rule policy over calibrated signals; it is implemented twice, in Python and TypeScript, and parity-tested. The engineering budget is zero: everything runs in one free container with a 512 MiB RAM ceiling on one deployment target, which excludes large local models and paid APIs alike.

The evaluation was disciplined for a hackathon: decision rules were fixed before results were seen, thresholds were chosen on validation only, a coverage target was corrected once measurement proved it unreachable rather than the system being bent toward it, and every displayed number is read from a single artifact (`backend/artifacts/eval.json`). The headline measurements are:

- the shipped linear engine reaches 0.820 accuracy and 0.816 macro-F1 on 806 held-out test messages;
- at the default threshold $\tau = 0.85$ the policy auto-handles 53.6% of messages at 94.2% accuracy on that slice;
- temperature scaling cuts expected calibration error from 0.200 to 0.049 (15 bins);
- the scam head attains 96.9% recall at a 0.25% false-alarm rate on ordinary shop messages, with the threshold fixed on validation beforehand;
- the full signal path runs at p50 3.6 ms / p95 4.6 ms on CPU.

Applied papers in this space usually report a single accuracy figure for a fine-tuned transformer. We believe (and the literature review in Section 6 will verify) three angles are under-explored for this setting: cost as a first-class evaluation axis, the conditions under which an LLM "second opinion" helps rather than harms, and the decomposition of coverage ceilings in deferral policies. The domain itself — code-switched Bangla commerce messaging — is under-served relative to English support traffic.

2 Working title and venue ladder

The main working title is *"Cheap and Honest: Cost–Quality Trade-offs for Calibrated Selective Automation of Code-Switched Bangla E-Commerce Messages."* Two alternatives are held for sharper framings: (i) *"Handling Half the Inbox at p95 $<$ 5 ms: Selective Prediction and Bounded LLM Deferral for Bangla Commerce Messaging"* (systems angle), and (ii) *"When Not to Trust the Judge: Confidence-Bounded LLM Second Opinions for Low-Resource Triage"* (analysis angle).

Tier	Example targets	Added work required	Horizon
Workshop	ACL/EMNLP-affiliated workshop on code-switching or low-resource NLP	Re-run notebook; related work; figures	1–2 weeks
Q2/Q3 journal	*Natural Language Engineering*; *Computer Speech & Language*	Multilingual baselines; seeds, CIs, AURC; frozen LLM cache	4–6 weeks
Q1 journal	*Expert Systems with Applications*; *Eng. Applications of AI*; *Information Processing & Management*	Human annotation with agreement statistics; OOD $n \geq 500$; full ablations; 20–30 citations	2–3 months

Table 1: Venue ladder. Quartiles change over time; rankings must be re-checked at submission time. The tiers differ mainly in data verification and baseline depth, not in framing.

3 Research questions

RQ1 **Cost–quality frontier.** Under a free-tier deployment budget, where do a character n-gram linear model, multilingual encoders, and LLM APIs sit on the accuracy–latency–memory Pareto frontier for six-class triage of code-switched messages?

RQ2 **Bounded LLM deferral.** Does an independent LLM second signal improve selective accuracy at fixed coverage, and does that improvement survive once the local model is well calibrated? When the two disagree, which one should be trusted as a function of local confidence?

RQ3 **Coverage ceiling.** How much of the non-automatable share of an inbox is structural (messages that ask for a person, messages the model is genuinely unsure about) versus improvable by a better model?

RQ4 **Confidence under shift.** How does per-script performance (Bangla, Banglish, English, mixed) and out-of-distribution behaviour interact with calibration, i.e. is the confidence score still trustworthy where the model is weakest?

4 Contributions claimed

- **C1 — Selective-automation analysis of a shipped policy.** A complete coverage–accuracy curve over $\tau \in [0.50, 0.99]$ computed with the real routing function, plus a ceiling decomposition showing that a *perfect* escalate oracle still caps coverage at 56.2% because 26% of messages are not classified confidently and 13% explicitly ask for a person. The ceiling argument turns a product limitation into an analytical result.
- **C2 — Bounded LLM deferral, measured.** On the test split, cases where the LLM and the local model disagree at top probability ≥ 0.90 favour the local model 96% to 23%. We use this to justify and formalise a veto design in which the LLM may only send a message to review, never decide alone, and only when the local model is uncertain; an unconditional veto is measured as the contrast condition and costs both coverage and accuracy.
- **C3 — Cost as an evaluation axis.** The same task measured across a 12 ms linear model, a 322M-parameter checkpoint (45.6 ms, $\geq$3 GiB host requirement with an observed out-of-memory failure below that), and a rate-limited free-tier LLM API — including the refusal threshold that keeps a 512 MiB host alive.
- **C4 (conditional) — Verified evaluation slice.** A human-re-annotated subsample with inter-annotator agreement, if the annotation plan in Section 5.1 is executed. Without it, C4 is dropped and the paper does not claim label quality beyond the source datasets.

5 State of the material

5.1 Data

- **D1** — `nafiullah/bangla-ecom-voice-decisions` [9], 7,032 rows mapped from 15 fine-grained intents to the six product classes; split 794 validation / 806 test after near-duplicate removal against test. Licenced non-commercial; the corpus is *largely machine-generated*, which is limitation 1 of the artifact's own limitations list and the main threat this proposal must answer.
- **D2** — `Piash2026/bengali-sms-smishing-dataset` [11], 7,005 SMS, of which the fraud classes are kept and marketing/propaganda labels excluded by a documented decision. SMS-style, not shop-chat — limitation 3.
- **D3** — BanglaTOD e-commerce subset [1], 105 customer turns, mapped from 10 intents to 6 classes with the mapping written out as explicit lines so a reader can disagree with a specific decision. Single-source, small, and judgement-dependent — limitation 4.

5.2 Measured results to date

All values from `eval.json` (2026-10-16); Engine A in that run was the character n-gram linear model.

Two attribution notes that any paper must keep: the 0.989 macro-F1 of the 322M-parameter checkpoint is an *in-distribution* number measured on the corpus the checkpoint was fine-tuned on (100 validation rows, gate rule fixed in advance), so it may headline a table but never a claim of generalisation; and the notebook currently stores no executed outputs — it is a narrated protocol, so the first week of any plan is a clean re-run that regenerates every number from pinned code.

5.3 What is missing for a journal-grade study

1. Baseline depth: no fine-tuned mBERT/XLM-R/Bangla-specific encoder rows, and only one hosted LLM for the zero-shot row.

System	Accuracy	macro-F1	Scoring subset
B1 linear engine, calibrated (E1)	0.820	0.816	all 806 test messages
B2 LLM alone (zero-shot)	0.616	0.520	LLM-answered subset[a]
E2 decision + LLM agreement rule	0.953	0.945	decided subset[a]
E3 full policy, $\tau = 0.85$	0.942	0.938	auto slice, cov. 53.6%
Calibration (ECE, 15 bins)	$0.200 \rightarrow 0.049$	—	before/after
Scam head	P 0.963 / R 0.969	F1 0.966	FAR 0.25% on D1
OOD (BanglaTOD)	0.362	—	$n = 105$

Table 2: Existing evidence. [a]E2 and B2 accuracy is computed on the messages each system chose to answer; the notebook records this caveat, and a paper must present these rows as risk–coverage pairs, never as directly comparable accuracies. Per-script accuracy: Banglish 0.881, Bangla 0.754, English 0.811, mixed 0.943 ($n = 35$).

2. Statistics: one split, no seeds, no confidence intervals, no significance tests; abstaining systems are compared on different subsets.
3. Label verification: no inter-annotator agreement anywhere; the escalate flag is a policy annotation, not ground truth.
4. OOD scale: $n = 105$ with a judgement-call mapping is a caveat, not a result.
5. LLM reproducibility: rows measured against a live API whose models rotate; the cache and model ids must be frozen before submission.
6. Related work: the project currently contains no citations at all.

6 Related work (to be written)

The final paper will organise prior work by approach, not chronology; the four strands below are the planned structure, each with the gap it must establish.

- **Calibration and selective prediction.** Platt scaling [12] provides the sigmoid correction lineage used for the binary heads, and Chow's reject-option work [3] is the origin of trading errors against abstention. Temperature scaling [7] established that modern networks need post-hoc scalar calibration, and selective-classification work [6] formalises accuracy-under-abstention, which is exactly the quantity our coverage curve reports; out-of-distribution behaviour is judged against detection baselines [8]. Our use is applied rather than methodological: the contribution is the measured behaviour of these tools on this task, not a new estimator.
- **LLM routing and cascades.** Judge-style evaluation [13] assumes the judge is reliable; routing and frugality work [2, 10] reduces cost by sending easy cases to small models and hard cases to large ones, usually with preference data or a learned router. Our setting has neither budget for preference collection nor labels for a router: the design under test is a *rule-based bounded veto* — the LLM may only disagree the system into review, only below a decisive-confidence gate — and we measure the contrast conditions (no veto, unconditional veto).
- **Multilingual and code-switched encoders.** Bidirectional pretraining [5] and its large-scale multilingual extension [4] underpin every candidate encoder row; Bangla-specific variants will be surveyed at execution time. The gap we test is per-script behaviour under a single deployed policy — one operating point, four script modes — rather than another pooled F1 score.

- **Bangla resources and fraud detection.** D1 [9] and D2 [11] are the task sources, and BanglaTOD [1] supplies the only out-of-source check available today; SMS-phishing literature will be surveyed for metric conventions. The established gap is domain mismatch (SMS versus shop chat) plus corpus provenance (machine-generated text), which Section 7 addresses by verification rather than by claim.

The gap paragraph will argue, subject to the review's outcome, that existing evaluations report pooled accuracy on pooled test sets, while the operational question — *what can be automated, at what confidence, at what cost, under a free-tier budget* — requires selective metrics, cost axes, and per-script slices reported together.

7 Proposed methodology

7.1 Data and verification plan

First, re-run the existing pipeline end to end from pinned code and regenerate `eval.json` with seeds recorded. Second, the annotation pilot: two annotators independently label a stratified subsample of at least 300 messages across class × script cells (including all low-confidence errors and a matched high-confidence sample), Cohen's κ is reported, disagreements are adjudicated, and the adjudicated slice becomes the *verified* test subset on which headline numbers are repeated. If κ is poor, that result is itself a finding about the source labels and is reported as such. Third, OOD expansion: at least 500 customer messages from at least two sources outside D1 with a mapping validated by the same two annotators; if the mapping proves unreliable, OOD stays qualitative, consistent with the existing evaluation plan. Datasets with non-commercial licences are used for non-commercial research only, and the licensing of any released subset is checked before publication.

7.2 Systems under comparison

Id	System	Status
A1	char n-gram TF-IDF + logistic regression (shipped)	exists
A2	mBERT-base fine-tune	to train, 5 seeds
A3	XLM-R-base fine-tune	to train, 5 seeds
A4	Bangla-specific encoder (survey)	to select
Z1	open instruct LLM, zero-shot, quantised locally	to run
Z2	hosted free-tier LLM, zero-shot (frozen ids + cache)	to freeze
L1	322M multilingual checkpoint (in-distribution caveat)	gate pass
E3	full policy: A1/L1 + escalate + scam + bounded LLM veto	exists

Table 3: Planned comparison. Every trainable row runs with five seeds; every row reports accuracy, macro-F1, latency, and peak RSS on the same machine.

7.3 Metrics and statistics

Classification: accuracy, macro-F1, per-class and per-script values, each with bootstrap 95% confidence intervals (10,000 resamples). Pairwise system comparisons use McNemar's test on identical subsets; seed-trained systems are reported as mean ± std. Selective prediction: coverage, auto-slice accuracy, and area under the risk–coverage curve, with all abstaining systems compared at matched coverage rather than at their own operating points. Calibration: ECE (15 bins), adaptive ECE, Brier score, and classwise ECE before and after temperature scaling, with intervals. Fraud detection: precision, recall, F1 on D2 positives against D1 shop-chat negatives, plus false-alarm rate with an interval and recall at a fixed 1% false-alarm rate. Latency and peak memory: p50/p95 on the deployment host, not the

workstation. Ablations: veto off / bounded / unconditional × a sweep of the decisive-confidence gate; escalate rule on/off; oracle-ceiling decomposition of coverage; the scam invariant (nothing at or above the scam threshold may enter the auto lane) tested as an assertion, not reported as a number.

7.4 Reproducibility package

An executed notebook with stored outputs, pinned environment files, the frozen LLM response cache with model ids and dates, seeds, figures regenerated from the notebook, and the evaluation script that produces `eval.json` from raw data in one command.

8 Experimental plan and timeline

Week	Work	Exit criterion
W1	Re-run pipeline + gate script; regenerate artifact; export figures	all numbers reproduce or differences are explained
W2	Statistical machinery: bootstrap CIs, McNemar, AURC, risk–coverage pairs	one table/format reused everywhere
W3–W4	Train A2–A4 (5 seeds each); select A4	baseline table complete
W5	Freeze Z2 (model ids, cache); run veto ablation grid	LLM rows reproducible offline
W6	Annotation pilot (300 msgs, κ); start OOD expansion	verified slice exists or κ finding reported
W7	OOD completion; analysis and error sections	RQ1–RQ4 each answered with a table
W8	Related work, figures, internal review; submit workshop version	submission-ready PDF

Table 4: Eight weeks to a workshop submission; the Q1 extension (full OOD, deeper ablations, expanded related work) continues through month 3 on the same skeleton.

9 Risks and mitigations

10 Honest limitations and compliance

The paper inherits the artifact's thirteen-item limitations list verbatim in substance: machine-generated intent corpus; machine-assisted gold labels; SMS-style fraud data; single-source in-distribution test with a small OOD check; simulated inbox with no live platform integration; class-granularity decisions; engine attribution for every reported number. Framing rules also carry over: the corpus is a *public Bangla e-commerce customer-message corpus*, never "real seller messages"; no accuracy figure is hardcoded anywhere in the product; and the bibliography in this proposal is provisional and will be verified field by field before submission. Authorship practice follows the project's rule that an AI coding harness did substantial work: the venue's AI-disclosure policy will be checked and complied with, and the contribution claims here are proposals for measurement, not measurements, until the re-run in W1 produces them.

11 Deliverables

(1) Workshop paper, weeks 1–8, on C1–C3 with the re-run evidence; (2) reproducibility package (executed notebook, cache, seeds, one-command artifact regeneration); (3) journal extension with the verified

Risk	Consequence	Mitigation
Corpus is machine-generated	Reviewer rejects label validity	Annotation pilot with κ; verified subset carries headline numbers; limitation stated first, not last
No novelty beyond application	Desk rejection at Q1	Anchor the paper on C1–C3 (ceiling analysis, bounded-veto result, cost frontier); if the review shows the angle is taken, retarget to a workshop
LLM rows drift or become paid	Central table not reproducible	Freeze model ids, ship response cache, re-measure dates; degraded rows allowed
Single-author capacity	Eight-week plan slips	Workshop tier is the fixed commitment; journal tiers are stretch goals
Non-commercial licences	Cannot release or reuse some rows	Use for research only; release only cleared subsets
The 0.989 checkpoint number leaks into claims	Overclaim; violates the project's honesty rules	It is reported as in-distribution evidence with its fine-tune provenance attached, never as a headline

Table 5: Top risks. The last row is a content risk, not a statistical one, and is governed by the project's honesty rules.

slice and expanded OOD, targeted at the Q2/Q3 or Q1 tier depending on what the data supports; (4) if and only if C4 survives annotation, release of the verified subset under cleared licences.

12 Conclusion

The proposal does not claim that the existing artifact is a paper; it claims that the artifact already contains three measurable contributions and an unusually honest evaluation protocol, and that six specific experiments plus statistical and human verification convert it into one. The gate for continuing is cheap: W1's re-run either reproduces the numbers or ends the project early.

References

[1] BanglaTOD: a multi-turn task-oriented dialogue dataset, 2022. TO VERIFY: full title, authors, year, venue, URL – placeholder entry, replace with the canonical citation.

[2] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and error. *arXiv preprint arXiv:2305.05176*, 2023. TO VERIFY: arXiv id.

[3] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.

[4] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2019.

[6] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[7] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[8] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.

[9] nafiullah/bangla-ecom-voice-decisions: a public bangla e-commerce customer-message corpus. `https://huggingface.co/datasets/nafiullah/bangla-ecom-voice-decisions`. TO VERIFY: licence text and access date.

[10] Ian Ong et al. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024. TO VERIFY: arXiv id and full author list.

[11] Piash2026/bengali-sms-smishing-dataset. `https://huggingface.co/datasets/Piash2026/bengali-sms-smishing-dataset`. TO VERIFY: licence text and access date.

[12] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999. TO VERIFY: publisher and exact year.

[13] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yu Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Joseph E. Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.